



PROJECT DESIGN REPORT

Grama-Angana: Community Hall Management Android App



A Project Report Submitted in Partial Fulfillment of the
Requirements for the Internship Project

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Grama-Angana: Community Hall Management Android App

Project Title: Grama-Angana (Infrastructure)

Project Type: Android Mobile Application

Platform: Android Studio (Kotlin / Java)

Database: Firebase Realtime Database / Firestore, Room Database

Authentication: Firebase Authentication

Prepared By: Chandrashekar

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1. Introduction

1.1 Project Overview

Grama-Angana is a Community Hall Management Android Application designed to transform village community halls (Anganas / Samudaya Bhavanas) into active hubs of social and cultural activity. The app addresses the core problem of underutilized public infrastructure by providing a transparent, digital platform for hall booking, maintenance tracking, and community crowdfunding.

1.2 Problem Statement

Community Halls (Anganas / Samudaya Bhavanas) in rural areas are often left empty and underutilized. Local youth and community groups want to use these spaces for sports, training, meetings, and events, but face critical information gaps:

- **No visibility** into the current booking status of the hall
- **No clarity** on who holds the keys or manages the space
- **No formal process** to request a booking slot
- **Lack of maintenance tracking**, leading to gradual decay of the building
- **No transparency**, which sometimes results in favoritism in allocating public spaces

Over time, this leads to the building falling into disrepair and the community losing a valuable shared asset.

1.3 Vision

Grama-Angana acts as a “**Community Space Manager.**” It turns the village hall into a “**Hub of Activity**” by:

- Providing a **public calendar** where anyone can see when the hall is free or booked
- Enabling **simple booking requests** to the Panchayat
- Allowing the community to **crowdfund small repairs** (e.g., “Need ₹500 for a new fan”)
- Displaying an **Event Board** showing what is happening today

2. Objectives & Goals

2.1 Primary Objectives

1. **Digitize Hall Booking:** Replace informal, paper-based, or verbal booking systems with a transparent digital calendar.
2. **Prevent Double Booking:** Ensure no two users can book the same time slot.
3. **Enable Maintenance Tracking:** Allow community members to report and pledge support for repair needs.
4. **Improve Transparency:** Make booking decisions visible to all, reducing favoritism.
5. **Increase Asset Utilization:** Ensure public buildings are used to their full potential.

2.2 Impact Goals

Goal	Description
Asset Utilization	Ensuring public buildings are used to their full potential instead of remaining idle.
Social Capital	Bringing people together for shared community goals such as health camps, training, and celebrations.
Transparency	Preventing favoritism in the booking of public spaces by making the process open and auditable.

2.3 Success Criteria

- The **Calendar** must prevent **Double Booking** of the same slot.
 - The **Maintenance Jar** must show a **progress bar** for funds / pledges.
 - The **UI** must be professional and community-focused.
 - The **Admin Dashboard** must allow approval / rejection of booking requests.
 - The **Event Board** must display today's events clearly.
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3. Scope

3.1 In-Scope (MVP)

The Minimum Viable Product (MVP) for this internship project includes:

1. **Hall Booking System** — Users can view availability and request slots.
2. **Calendar Tracking** — Visual calendar showing booked vs. free slots.
3. **Admin Approval Workflow** — Panchayat admin can approve or reject requests.
4. **Maintenance Requests** — Users can report repair needs and view the maintenance list.
5. **Basic Donation / Pledge Tracking** — Users can pledge support for repairs with a visible progress bar.

3.2 Out-of-Scope (Future Enhancements)

The following features are identified for future releases and are excellent discussion points for interviews:

1. **Online Payment Integration** — UPI payments and booking fees.
2. **QR Entry System** — Generate QR code after booking; verify at hall entrance.
3. **AI-based Demand Prediction** — Predict busy seasons and suggest available dates.
4. **Notification System** — Booking reminders and event alerts via push notifications.
5. **Multi-Hall Support** — Manage multiple village halls from a single app.
6. **Google Maps Integration** — Hall location and navigation.
7. **Document Upload** — Upload ID proof for booking verification.
8. **Analytics Dashboard** — Most booked months, revenue reports, usage statistics.
9. **Chat Support** — Direct communication with the hall manager.
10. **Complaint System** — Report damaged equipment with photo evidence.

4. Functional Requirements

4.1 Module 1: User Login / Register

ID	Requirement	Priority
FR-1.1	Users can register with email / phone and password.	High
FR-1.2	Users can log in using Firebase Authentication.	High
FR-1.3	Admin / Panchayat members have a separate login with elevated privileges.	High
FR-1.4	Password recovery option must be available.	Medium

4.2 Module 2: Hall Availability Calendar

ID	Requirement	Priority
FR-2.1	CalendarView must display the current month with date-wise status.	High
FR-2.2	Booked slots must be visually marked (e.g., red or shaded).	High
FR-2.3	Free slots must be visually marked (e.g., green).	High
FR-2.4	Users can tap a date to see detailed slot information (morning / evening).	High

4.3 Module 3: Booking Request System

ID	Requirement	Priority
FR-3.1	Users can select a date and time slot from the calendar.	High
FR-3.2	Users must fill a simple form with event type and purpose.	High
FR-3.3	The request is sent to the admin for approval.	High
FR-3.4	Users can view the status of their booking requests (Pending / Approved / Rejected).	High

4.4 Module 4: Double Booking Prevention

ID	Requirement	Priority
FR-4.1	The system must check slot availability in real-time before confirming a request.	High

FR-4.2	If a slot is already booked or pending approval, it must be blocked for new requests.	High
FR-4.3	Only after admin approval must the slot be marked as “Booked.”	High

4.5 Module 5: Event Board

ID	Requirement	Priority
FR-5.1	A dedicated screen must show today’s events.	High
FR-5.2	Events must be categorized: Wedding, Meeting, Training, Sports, Health Camp, etc.	Medium
FR-5.3	Event details must include time, organizer name, and event type.	Medium

4.6 Module 6: Maintenance Module

ID	Requirement	Priority
FR-6.1	A list of repair needs must be displayed (e.g., fan repair, broken chairs, bulb replacement).	High
FR-6.2	Users can view estimated cost for each repair item.	Medium
FR-6.3	Users can pledge support (monetary or in-kind) toward a repair item.	High
FR-6.4	A progress bar must show how close each repair item is to its funding target.	High

4.7 Module 7: Crowdfunding / Pledge System

ID	Requirement	Priority
FR-7.1	Each maintenance item must have a target amount.	High
FR-7.2	Users can pledge an amount toward any item.	High
FR-7.3	A visual progress bar must update dynamically as pledges are made.	High
FR-7.4	Total pledged amount and remaining amount must be visible.	Medium

4.8 Module 8: Admin Dashboard

ID	Requirement	Priority
FR-8.1	Admin can view all pending booking requests.	High
FR-8.2	Admin can approve or reject each request with a single action.	High
FR-8.3	Admin can update hall status (e.g., temporarily closed for maintenance).	Medium
FR-8.4	Admin can add or edit maintenance items and their target amounts.	Medium
FR-8.5	Admin can view a summary of bookings, pledges, and hall usage.	Medium

5. Non-Functional Requirements

ID	Requirement	Priority
NFR-1	The app must load the calendar within 3 seconds.	High
NFR-2	Booking requests must be processed in real-time using Firebase.	High
NFR-3	The UI must be intuitive for users with limited digital literacy.	High
NFR-4	The app must support offline viewing of the calendar (cached data).	Medium
NFR-5	Data must be secure and accessible only to authenticated users.	High
NFR-6	The app must be responsive across different Android screen sizes.	Medium

6. User Flow & App Usage

6.1 General User Flow

1. **Launch App** → Login / Register screen
2. **Home Screen** → View Event Board (today's events) and quick actions
3. **Calendar Screen** → View booked vs. free slots; tap a date to request booking
4. **Booking Form** → Fill event details and submit request
5. **Maintenance Jar** → View repair needs; pledge support
6. **My Bookings** → Track status of submitted requests

6.2 Admin / Panchayat User Flow

1. **Admin Login** → Access elevated dashboard
 2. **Pending Requests** → Review and approve / reject booking requests
 3. **Calendar Management** → Update hall status, mark holidays
 4. **Maintenance Management** → Add new repair items, update funding targets
 5. **Reports** → View summary statistics (optional for MVP)
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7. Technical Implementation

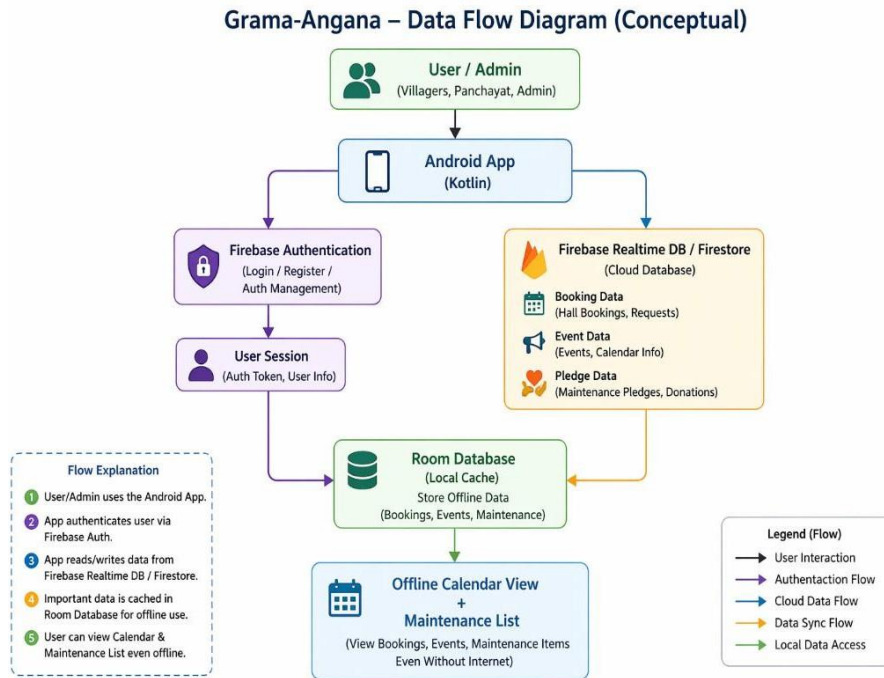
7.1 Technology Stack

Layer	Technology	Purpose
IDE	Android Studio	Development environment
Language	Kotlin / Java	Native Android development
Authentication	Firebase Authentication	User login, admin login, secure access
Real-time Database	Firebase Realtime Database / Firestore	Booking requests, event data, live updates
Local Database	Room Database	Offline storage of maintenance items, cached calendar data
UI Components	CalendarView, RecyclerView, ProgressBar	Calendar display, lists, funding progress

7.2 Technical Hints & Architecture

- **UI:** Use a CalendarView with custom status markers (colors or badges) to indicate booked, free, and pending slots.
- **Real-time:** Use Firebase for booking requests so that admins and users see updates instantly without refreshing.
- **Database:** Use Room DB to store the list of maintenance items locally for fast access and offline capability.
- **Conflict Handling:** Implement server-side validation (Firebase security rules or Cloud Functions) to prevent race conditions and double booking.
- **Progress Tracking:** Use ProgressBar UI component bound to pledge data from Firebase.

7.3 Data Flow Diagram (Conceptual)



8. System Architecture

8.1 High-Level Architecture

The application follows a **client-server architecture** with local caching:

- **Presentation Layer:** Android Activities / Fragments with XML layouts
- **Business Logic Layer:** ViewModels, Repositories, Use Cases
- **Data Layer:** Firebase (cloud) + Room Database (local)
- **Authentication Layer:** Firebase Authentication

8.2 Key Android Components

Component	Usage
CalendarView	Display monthly calendar with slot status
RecyclerView	List of events, maintenance items, booking requests
ProgressBar	Visual indicator for crowdfunding targets
FirebaseAuth	User and admin authentication
FirestoreDatabase / Firestore	Real-time cloud data storage
Room	Local SQLite abstraction for offline data

9. Database Design (Conceptual)

9.1 Firebase Collections / Nodes

users - userId, name, email, phone, role (user / admin), villageId

bookings - bookingId, userId, date, timeSlot, eventType, purpose, status (pending / approved / rejected), createdAt

events - eventId, bookingId, title, type, date, timeSlot, organizer, description

maintenanceItems - itemId, title, description, estimatedCost, pledgedAmount, status (open / in-progress / completed), createdAt

pledges - pledgeId, itemId, userId, amount, pledgedAt

9.2 Room Database Entities (Local Cache)

MaintenanceItemEntity - itemId, title, description, estimatedCost, pledgedAmount, status

CachedBookingEntity - bookingId, date, timeSlot, status, eventType

10. UI / UX Design Considerations

10.1 Design Principles

- **Community-Focused:** Use warm, welcoming colors and simple iconography.
- **Low Digital Literacy Friendly:** Large buttons, clear labels, minimal text input.
- **Visual Clarity:** Color-coded calendar (green = free, red = booked, yellow = pending).
- **Local Language Support:** Consider Kannada / regional language labels for broader adoption.

10.2 Key Screens

1. **Splash / Login Screen**
2. **Home / Dashboard** — Event Board + Quick Actions
3. **Calendar Screen** — Monthly view with slot markers
4. **Booking Request Form**
5. **Maintenance Jar Screen** — List with progress bars
6. **My Bookings Screen** — Status tracking
7. **Admin Dashboard** — Pending requests, hall status, maintenance management

11. Real-World Impact

11.1 Community Benefits

Impact Area	Description
Better Use of Public Buildings	Halls are no longer left idle; they become active centers for learning, health, sports, and celebrations.
More Transparency	Booking decisions are visible to all villagers, reducing disputes and favoritism.
Prevent Favoritism	Digital records ensure fair allocation of public spaces on a first-come, first-served or merit basis.

Helps Community	Village	Strengthens social bonds by enabling shared activities and collaborative maintenance.
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11.2 Why This Project Stands Out

This is **not just a CRUD application**. It includes:

- **Booking Logic** — Complex date / time slot management
- **Conflict Handling** — Real-time double booking prevention
- **Admin Workflow** — Approval / rejection with state management
- **Fund Tracking** — Crowdfunding with dynamic progress visualization
- **Real-World Scalability** — Designed to support multiple halls, payments, and analytics in the future

These dimensions make it a strong portfolio project for internships and placements.

12. Project Timeline (Suggested)

Phase	Duration	Activities
Phase 1: Setup & Design	Week 1	Android Studio setup, Firebase project, UI wireframes, database schema design
Phase 2: Authentication & Calendar	Week 1	Firebase Auth, CalendarView integration, slot status display
Phase 3: Booking System	Week 2	Booking request form, double booking prevention, admin approval workflow
Phase 4: Maintenance & Pledges	Week 2	Maintenance list, pledge system, progress bar UI, Room DB integration
Phase 5: Event Board & Polish	Week 3	Event board, UI refinement, testing, bug fixes
Phase 6: Documentation & Demo	Week 3	PDR report, user manual, demo video, presentation

13. Risk Analysis

Risk	Impact	Mitigation
Internet connectivity issues in rural areas	High	Implement Room DB for offline calendar viewing and sync when online
Low smartphone penetration among elderly	Medium	Design simple UI; encourage youth to assist elders
Firebase free tier limits	Low	Monitor usage; optimize data structure; upgrade if needed
Resistance to digital booking from Panchayat	Medium	Demonstrate transparency benefits; provide admin training

Device fragmentation (screen sizes) Low Use responsive layouts and ConstraintLayout

14. Testing Strategy

Test Type	Focus Area
Unit Testing	ViewModels, Repository logic, booking conflict detection
Integration Testing	Firebase Auth flow, Room DB operations, Firebase sync
UI Testing	Calendar navigation, booking form submission, progress bar updates
User Acceptance Testing	Test with actual villagers and Panchayat members for usability feedback

15. Conclusion

Grama-Angana is a meaningful, technically robust Android project that solves a real problem in rural India. By digitizing hall booking, enabling transparent maintenance crowdfunding, and providing a community event board, the app transforms underutilized public buildings into vibrant community hubs.

The project goes beyond basic CRUD operations by incorporating:

- Real-time conflict resolution
- Admin workflow management
- Dynamic fund tracking
- Scalable architecture for future enhancements

This makes it an excellent choice for an internship project — one that demonstrates technical competence, social awareness, and product thinking.

16. References & Resources

- Android Developer Documentation: <https://developer.android.com/>
- Firebase Documentation: <https://firebase.google.com/docs>
- Room Persistence Library: <https://developer.android.com/training/data-storage/room>
- Material Design Guidelines: <https://m3.material.io/>